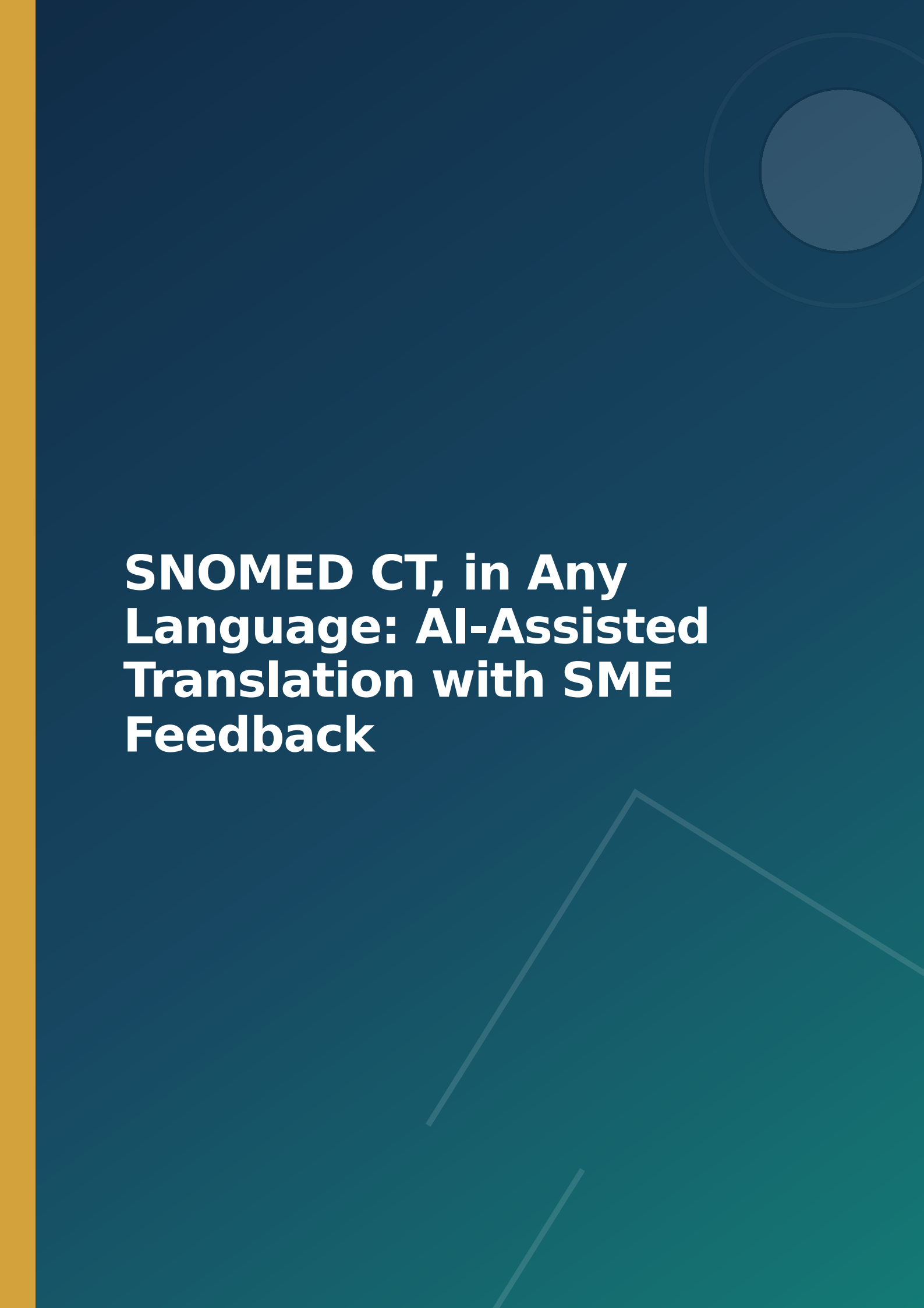




SNOMED CT, in Any Language: AI-Assisted Translation with SME Feedback

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Executive summary

Every SNOMED CT language project starts from a different position. One country may have a substantial but incomplete national extension; another may have only glossaries, mappings, clinical writing, and a small group of experts. Language rules also differ across clinical domains. A prompt that works for Korean imaging is not automatically suitable for Norwegian findings or Spanish substances.

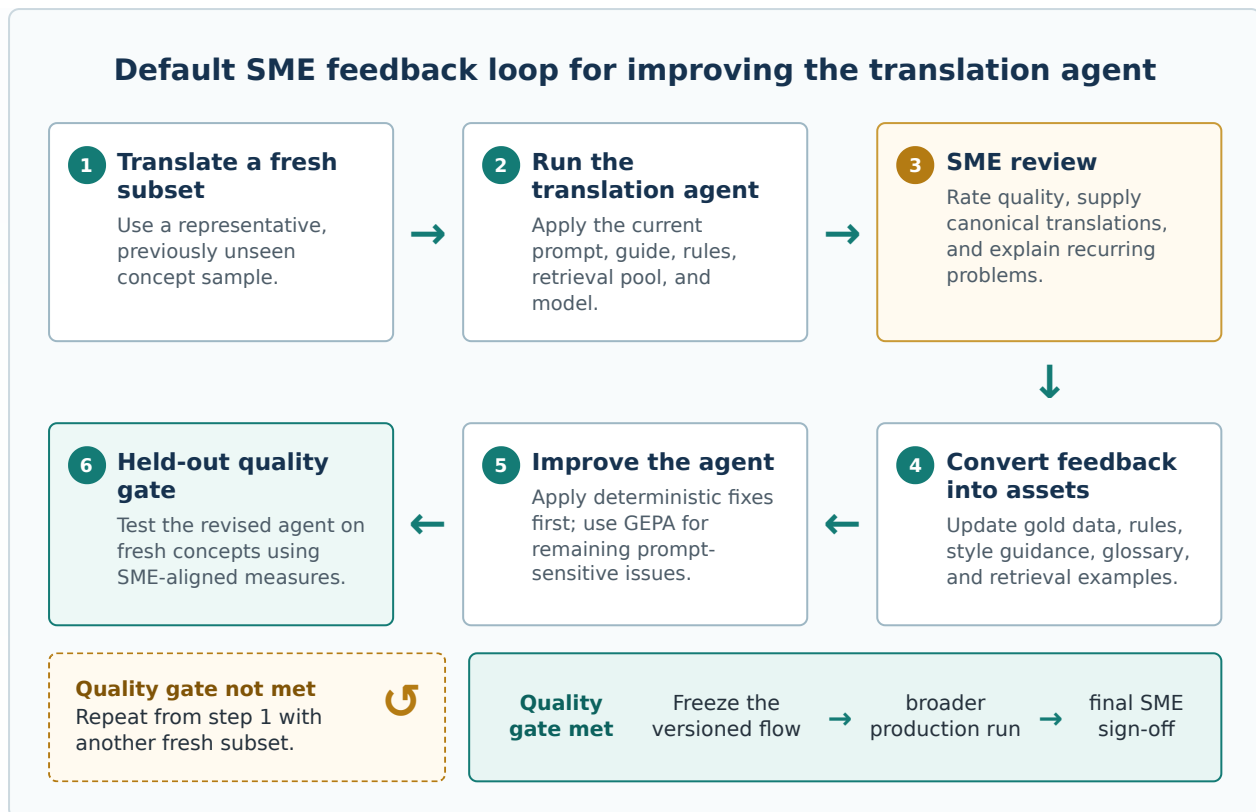
The platform is therefore built around a simple proposition:

What transfers between languages is not one fixed prompt. It is a measured, auditable process for finding and validating the right translation approach.

The SME feedback loop shown below is the platform's **default template and recommended starting point, not a fixed translation algorithm**. Treating each language as a research programme means the team can investigate a different or more complex end-to-end process when evidence supports it. Experiments may change the agentic design—for example, using separate drafting, critique, or adjudication roles—as well as model routing, the selection and presentation of supporting context, hierarchy-specific information, evaluators, quality checks, and repair stages. Prompt evolution through GEPA is one available optimisation method rather than the only kind of improvement.

The supplied default begins by importing existing resources and indexing them for semantic search. A base translation guide seeds the agent. A representative subset is translated and reviewed by a terminology subject-matter expert (SME). The SME supplies canonical translations and quality ratings; those decisions become adjudicated gold data, rules, style-guide improvements, and better retrieval examples. Whatever process is investigated, competing flows are compared through tracked runs and held-out, SME-aligned evidence before one is promoted.

The cycle repeats on fresh concepts until the agreed quality gate is met. Only then is the winning, version-pinned flow used at broader scale. The final output remains subject to SME review and sign-off.



The capability can be made available in two complementary ways:

- an **open-source, self-hosted edition**, where customers receive the project and connect their own local or API models; and
- a **managed service**, where we operate the platform and supply cloud inference, charging for onboarding, platform access, usage, optional optimisation, and any human services.

1. The capability in one view

The workbench combines six functions that would otherwise be separate pieces of work:

1. **Project setup** — the reusable `translation_process` template installs the plan, gates, flows, starter rules, source definitions, and methodological safeguards.
2. **Resource ingestion** — SNOMED CT content and available language resources are registered with their provenance and licence conditions.
3. **Semantic retrieval** — bilingual pairs and other approved contextual material are indexed so the translation agent sees relevant examples at translation time.
4. **Controlled translation research** — models, prompts, context construction, agent roles, routing, quality checks, parameters, datasets, and whole flow designs are explicit and versioned.
5. **SME feedback and optimisation** — review creates canonical gold data and editorial rulings; the evidence can improve prompts, retrieval, rules, evaluators, or the wider agentic process.
6. **Quality assurance and delivery** — automated detectors, blinded samples, audit trails, promotion gates, and SME sign-off protect the release.

Every material number should come from a tracked run. Every promoted result should identify the source data, code, model, prompt, style guide, and rules that produced it.

2. What is reusable and what each language supplies

Reused for every language	Supplied or developed for each language
Project template, plans, gates, and workflow engine	Licensed SNOMED CT release and chosen hierarchy scope
Import and semantic-indexing machinery	Existing extension, translation pairs, glossaries, mappings, or suitable clinical narrative
Model registry for local and API endpoints	A base translation guide and target-language conventions
Translation, evaluation, QA, review, and repair flows	SME-adjudicated translations for optimisation and held-out evaluation
GEPA prompt-optimisation workflow	Terminology SME time and authority to set the quality bar
Run tracking, provenance, and audit trail	Language-specific rules, glossary, and curated retrieval pool as they mature
Tool interface through which an AI assistant can operate the research workflow	Decisions on licensing, ownership, residency, publication, and maintenance

This separation is why the platform can scale. The engineering method is reused; the language receives the data and expert attention it genuinely needs.

Different parts of SNOMED CT may also be treated as separate research scopes. Procedures, findings, substances, and organisms follow different naming conventions. A smaller guide, exemplar pool, and gold set for one hierarchy section can outperform one oversized prompt covering every domain. Sections can be researched and promoted on different timetables, beginning where clinical value is highest.

3. The end-to-end language process

Step 1 — Define and provision the project

Choose a coherent hierarchy section or clinical domain rather than beginning with the whole terminology. Record the SNOMED CT edition, release date, modules, relationships, target language, model configuration, and intended publication route.

Seed the project from `translation_process`. This installs the default SME feedback workflow as a working reference that the research team can clone, extend, or replace with an evidence-backed alternative. Its first gate remains blocked until the terminology, usable translation resources, SME commitment, and domain slice are available.

Step 2 — Bring the available language resources

Useful starting material may include an incomplete national extension, approved translation pairs, terminology mappings, medical dictionaries, standard vocabularies, or representative clinical writing. Each source must have suitable reuse rights and clear provenance.

Translation pairs are curated and indexed semantically. When the agent receives an English concept term, it retrieves the closest approved examples and includes them with the style guide, applicable rules, and available SNOMED CT attributes. This gives the model language-specific context without treating the most frequent wording as automatically canonical.

A greenfield language needs an explicit seed-pool activity. An empty exemplar pool is an honest starting state, but it is not enough for a retrieval-assisted production run.

Step 3 — Establish a baseline

Supply a draft translation guide. It can be short: script, spacing, abbreviations, format, and known terminology policy. Detailed rules should emerge from SME evidence rather than being guessed in advance.

Translate a representative pilot subset spanning hierarchy positions, structural complexity, and expected difficulty. Run automated checks for rule violations, lost modifiers, suspicious transliteration, hierarchy inconsistency, and collisions where distinct concepts receive the same translation.

The model is swappable. A local open-weight model can perform the routine bulk work; uncertain concepts can be routed to a stronger commercial endpoint. Because every configuration is scored, the local/API choice is an evidence-based cost and quality decision rather than a fixed architectural assumption.

Step 4 — Obtain structured SME feedback

Package a blinded, stratified sample—typically about 100–120 concepts—and ask the SME to provide:

- a rating such as perfect/canonical, acceptable, partial, or wrong;
- the canonical translation where correction is needed;
- accepted synonyms where useful; and
- answers to a small number of cross-cutting adjudication questions.

The SME remains the authority. Machine output never overwrites an adjudicated row. If a newer ruling changes an older decision, supersession is recorded and the affected rows are shown for confirmation.

Step 5 — Turn review into reusable language assets

Ingest the returned review through a tracked flow. Feedback is encoded in the most durable form available:

1. canonical terms and accepted variants enter the adjudicated gold set;
2. deterministic requirements become hard rules and safe repairs;
3. editorial and word-order guidance updates the style guide;
4. unsuitable or superseded examples are removed from future retrieval; and
5. open questions and supersessions remain visible in the adjudication record.

Re-translating the reviewed concepts shows whether the system has absorbed the decisions. Generalisation is tested on fresh held-out concepts, not on the rows used to develop the changes.

Step 6 — Optimise the prompt where appropriate

GEPA can propose and evaluate revised instructions using the growing set of trusted translations. The evolved guide remains a readable, versioned document that a human can accept, amend, or reject.

GEPA should be used after straightforward rules, glossary decisions, and pool corrections have been applied. Before a candidate prompt is promoted, the project needs:

- separated optimisation and held-out evaluation data;
- retrieval controls preventing a concept from seeing its own translation;
- a metric shown to align with SME judgement;
- frozen rules that candidates cannot trade away; and
- evidence that the candidate beats the current prompt on held-out concepts.

Step 7 — Repeat, then run at scale

Translate a fresh subset and repeat the review-improvement cycle. Early rounds normally settle high-volume language conventions; later rounds concentrate on difficult structure, rare terminology, and ambiguity.

Once the agreed quality threshold is met, freeze the winning model, prompt, guide, rules, pool, and code versions. Run the broader scope, audit the output, resolve blockers, and obtain formal SME sign-off before publication.

Two delivery patterns

The scope of the country's need determines how this process is applied:

- A **targeted-domain project** can run the complete feedback and improvement cycle for one clinical area, produce a signed-off domain translation, and stop there. A country does not need to commit to translating the wider terminology.
- A **broad-terminology programme** should avoid treating every domain as a separate project. Establish the language once, optimise an initial approach on a representative cross-domain sample, and then score its quality by domain. Repeat domain-specific SME feedback and optimisation only where a domain falls below the agreed quality gate. Domains that already meet the gate can proceed to production and proportionate assurance using the shared language configuration.

This staged approach creates potential economies of scale without assuming that one prompt will work equally well everywhere. The amount of domain-specific work remains an evidence question: it depends on cross-domain performance, clinical risk, and the level of assurance required by the country.

4. Evidence from the Korean work

The presentation [snomed-translation-deck.html](#) records an accessible early result: in the first 100-term SME review, 47 translations were rated acceptable, 51 partial, and two wrong. This meant 98 were usable or near-usable, but the 51 partial terms still required expert correction. The result demonstrated drafting value, not publication readiness.

The later project work strengthened both the method and the evidence:

- a 5,012-concept imaging domain was translated through a confidence-routed cascade;
- approximately 45% of concepts were escalated to the stronger model;
- the full run completed without errors and the local sampling stage took about 23 minutes on one GPU workstation;
- systematic SME rulings were converted into guide changes and deterministic rules, sharply reducing recurring convention defects; and
- blinded review, leakage-safe evaluation, explicit supersession, pair-level collision checks, and held-out promotion gates were added after weaknesses were found in earlier experiments.

The latest optimisation and feedback-integration round has now been incorporated into the translation configuration, and a further blinded SME review is pending. That review will assess fresh, previously unseen concepts and provide stronger evidence of how much the changes improve translation quality in practice. Until the feedback is returned, the impact of the latest optimisation round should be treated as promising but not yet established.

5. Cost and resource shape

Compute is measurable and comparatively inexpensive. Expert review is the scarce resource.

Machine cost

Prompt caching is the main API cost control for this workload and must be included in the headline estimate. The measured 5,012-concept production pass contained 16.87 million input tokens and about 33,000 output tokens. For planning purposes, assume that production prompt design and batching achieve a **90% cache-hit rate**:

Billing class	Volume at 90% cache	Current Qwen 3.8 Max international list rate	Estimated cost
Uncached input	1.687M tokens	\$2.00/M	\$3.37
Implicit-cache input	15.183M tokens	\$0.25/M	\$3.80
Output	0.033M tokens	\$6.00/M	\$0.20
Full-pass planning estimate			about \$7.37

Those rates were checked on 28 August 2026. The 90%-cached estimate is about 78% below the \$33.94 all-uncached calculation. Actual billing will depend on the achieved cache-hit rate, so the run dashboard should report it alongside the cost.

Qwen 3.8 Max is used here as a state-of-the-art frontier reference model. It is an API model rather than an open-weight release. The platform can instead use cheaper API endpoints or open-weight models running locally, including the Gemma-class model already used for bulk sampling. A local deployment has no per-token API charge, although its hardware and electricity remain operating costs.

Future quotations should use the run's recorded uncached, cache-read, cache-creation, and output tokens separately. Stable material—system instructions, the style guide, rules, and other repeated context—belongs at the beginning of the prompt so that the provider can reuse the longest possible prefix.

A project may need two or three broad passes, but small pilot and absorption runs cost a fraction of a full pass and deterministic repair requires no model call. API inference should therefore be presented as a small, measured operating cost; SME and service effort remain the material costs.

An open model can handle bulk sampling on owned or cloud GPU capacity. Semantic indexing is mainly an onboarding cost, and unchanged embeddings can be reused when the pool is updated. Deterministic QA and repair often require no model call.

GEPA evaluates many prompt candidates and should have a separate capped budget. Its proposed maximum should be calculated from candidate count, evaluation-set size, model routing, and cache-aware token rates before the run begins. Start with the smallest useful experiment, record actual cache performance and spend, and expand only when the quality gain justifies it.

Human resource

Activity	How the effort scales
Language and platform setup	approximately one operator week, primarily once per language—not once per domain or per 5,000 concepts
Initial optimisation	focused on the requested domain for a targeted project; based on a representative cross-domain sample for a broad programme
One standard feedback round	about 2–4 SME hours for a 100–120-term sample, plus approximately 1–2 operator days to prepare, integrate, and verify the round
Subsequent domain assessment	automated evaluation and sampled SME review before deciding whether more domain-specific work is justified
Domain-specific optimisation	conditional; repeat the feedback cycle only for domains that do not meet the shared quality gate
Targeted-domain planning anchor	about 10–20 SME hours across commonly 3–5 rounds for an imaging-scale domain with useful existing reference data
Greenfield allowance	a targeted project may need roughly 50% more SME effort and one or two additional rounds

The numerical anchors come primarily from the Korean imaging work. They describe a targeted domain of roughly 5,000 concepts; they are **not a unit cost to multiply across the full terminology**. A credible whole-terminology estimate requires the initial cross-domain evaluation to show how many domains meet the shared quality gate, how many need further optimisation, and how much sampled or exhaustive SME assurance each domain requires.

A customer price must also recover hosting, storage, indexing, monitoring, support, security, data governance, failed runs, provider-price changes, invoicing, and service margin.

6. Ways to make it available

Open-source, self-hosted

Customers receive the complete engine, project template, plugin interfaces, and deployment guidance. They provide their licensed terminology and language assets, deploy the vector database and storage, and connect any compatible local or API models.

This option supports strict residency and maximum technical control. We avoid inference-cost and credit exposure, but still need documentation, release management, compatibility testing, and a support model. Paid onboarding or enterprise support can sit around the open-source edition.

Managed service with our inference

We host and operate the workbench, semantic index, storage, monitoring, and approved model endpoints. The customer supplies its licensed content and SME access. We pay cloud and inference providers, meter usage, and bill the customer.

A practical charging structure is:

- onboarding fee
- + platform subscription with an included usage allowance
- + translated-concept or project-round overage
- + optional capped GEPA optimisation
- + professional or SME services where supplied

Raw tokens should be metered internally, but a concept- or round-based customer unit is easier to budget and better reflects the value of the workflow, QA, provenance, and support. Pricing needs contingency for model-price volatility and failed or repeated runs.

A managed customer could alternatively connect its own approved model endpoint. This helps with procurement or cloud policy but requires a clear shared-support boundary.